PHYSICAL MOUNTING SPECIFICATION

SMARTSTOCK SYSTEM • CONNECTED CABINET & MINIMAL EMERGENCY BAG

STATUS: **VALIDATED**
STOCK: **100% AVAILABLE**

CONNECTED CABINET	EMERGENCY BAG	FIXED CONTROLLER	BAG CONTROLLER
50x50x60 cm	**45x30x25 cm**	**Raspberry Pi 4**	**Arduino 1010**
Fixed Clinic Module	Waterproof 1680D Nylon	Dedicated Sliding Rail	Bluetooth + Battery

1. HARDWARE ARCHITECTURE OF PROTOTYPES

SPEC-01

MEDICAL CABINET (CLINIC)

FIXED

- **Structure:** Aluminum / ABS, vertical polycarbonate door with steel gas struts, removable shelves.
- **Weight Sensors:** Strain gauge sensors mounted beneath each shelf for continuous stock measurement.
- **NFC Identification:** Right internal NFC receiver (tracks bottles/medication on entry and exit).
- **Environment:** Phidget Temperature/Humidity sensor + extraction fan + automatic interior LED strip.
- **Control & Power:** Raspberry Pi 4 mounted on rear sliding rail, powered via 5V / 3A USB hub.

EMERGENCY BAG (VISITS)

PORTABLE

- **Structure:** Waterproof 1680D Nylon fabric, modular dividers, rigid anti-slip base.
- **Weight Sensors:** Individual weight sensors under each removable compartment.
- **NFC Reader:** Integrated beneath internal wall for instant tracking of additions/removals.
- **Bluetooth & Battery Module:** Protected compartment with waterproof USB-C external charging.
- **Additional Sensors:** Force sensor, vibration sensor, and automatic lid LED lighting.

2. HARDWARE INVENTORY ALLOCATION (SS-005.1)

MAP-01

SUBSYSTEM	CONNECTED CABINET (CLINIC)	MINIMAL EMERGENCY BAG (PORTABLE)
Central Unit	Raspberry Pi 4 + SD Card + Case + Heatsink	Genuino / Arduino 1010 + Bluetooth Module
Acquisition & Bus	Phidget VINT Hub + InterfaceKit 8/8/8	Prototyping board + Sensor box
Stock Sensors	Weight sensors (shelves), NFC, Barometer, Phidget Temp/Humidity	Weight sensors (compartments), NFC, Force sensor, Vibration sensor
Interface & Actuators	Graphic LCD Phidget, Joystick, Dual Relay Board, LEDs, Buttons	Touch sensors, Sliders, Status LEDs (Green/Red)
Power Supply	Dedicated 5V/3A Power Adapter + Powered USB Hub	Rechargeable USB-C PowerBank Battery

3. FEASIBILITY CONFIRMATION (NO EXTRA PURCHASE REQUIRED)

CHK-01

Requirement Validation: 100% of the components required to assemble the **Connected Cabinet** and **Emergency Bag** are covered by the existing inventory (SS-005.1). No supplementary purchases are needed. Electrical safety rules (SS-005.2) are fully respected via the isolated USB hub and low-voltage 3.3V controllers.